Response to Reviewer Comments

Manuscript Title:
Tracing Financialisation in Critical Metal Markets: Correlations, Bubbles, and Dynamic Linkages

August 26, 2026

We are grateful to the Editor and the Reviewers for their careful reading of our manuscript and for the opportunity to revise it. We believe the comments have led to substantial improvements. Below, we reproduce the reviewer comments *in italics*, followed by our responses in normal type. Passages quoted directly from the revised manuscript are set in blue.

Reviewer #1

The authors examine the existence of speculative bubbles and, in this context, employ a series of unit root tests followed by the Johansen cointegration approach to investigate the long-run relationships among the variables. While the topic is relevant, the study remains largely descriptive in nature and primarily reports whether certain statistical properties are present in the data. As a result, the manuscript lacks a sufficiently deep theoretical and empirical focus. In this regard, the authors should first report comprehensive descriptive statistics, as these may provide important insights into the characteristics of the data and help justify the adoption of additional models and analytical approaches. Subsequently, it should be demonstrated more convincingly why the analysis of speculative bubbles and long-run relationships constitutes the central focus of the study. Additional analyses are necessary, and bubble detection should not represent the entirety of the empirical contribution. Instead, the theoretical and empirical scope of the manuscript should be expanded. To this end, I offer the following recommendations.

Reviewer Comment:
A clearer alignment should be established between the identified gaps in the literature, the research question, the hypotheses, and the empirical framework.

Author Response:
We thank the Reviewer for this important observation. We agree that the previous version did not make the logical chain connecting the literature gap, research question, and empirical strategy sufficiently explicit, and that no formal hypotheses were stated. We have added a closing paragraph to the Introduction linking the research questions to the empirical strategy, and a new discussion at the end of the Literature Review that derives four testable hypotheses directly from three specific literature gaps.

Reviewer Comment:
The theoretical and empirical foundations of the hypotheses should be developed more

rigorously, and the hypotheses should be formulated as formal scientific statements.

Author Response:
We agree that hypotheses should be stated as formal, falsifiable propositions. We have formulated four such hypotheses at the end of the Literature Review, each grounded in a specific theoretical mechanism and directly falsifiable using the tests already employed in the manuscript:

> H1: The intensity of speculative bubble behaviour across the sixteen price series is systematically related to their relative illiquidity, as proxied by the incidence of zero-return trading days.
>
> H2a: Filtering out detected speculative bubble episodes strengthens the long-run cointegrating relationship among a metal's price series.
>
> H2b: The strength of long-run price cointegration across metals reflects their relative level of financialisation.
>
> H3: Short-run conditional correlations are markedly weaker between the two thinly traded metals (lithium, cobalt) and the two liquid metals (copper, nickel) than between copper and nickel themselves.

Each hypothesis is preceded by a short passage setting out the mechanism it tests and the literature that motivates it; H2a and H2b share one such passage.

Reviewer Comment:
The manuscript should clearly identify which specific research question cannot be adequately addressed by existing studies and explain why the combination of methods employed in the study generates fundamentally new insights.

Author Response:
We thank the Reviewer for pushing us to sharpen this point. As documented in our revised Literature Review, existing studies on critical-metal bubbles are almost exclusively single-metal and single-method, and studies of financialisation based on cointegration or connectedness do not control for speculative bubbles when estimating long-run relationships. We have added a direct comparison to the closest related study:

> Closest in scope to the present study, Khurshid et al. (2026) examine time-frequency return connectedness between several green minerals - copper, cobalt, lithium and nickel among them - and a mix of clean and fossil energy assets, finding connectedness strongest at short horizons and diminishing over longer ones. Their framework, however, treats each mineral as a single series and focuses on cross-asset-class spillovers with the energy sector; it does not address whether price series representing the same metal from different exchanges or benchmarks converge to a common long-run equilibrium, which is the specific question this study investigates.

No existing study, to our knowledge, jointly examines bubbles, bubble-filtered long-run cointegration, and short-run DCC dynamics for the same set of assets. The combination is more than additive: a low cointegration rank on raw prices is observationally consistent with both genuine structural fragmentation and bubble-masked equilibrium, and the raw-versus-filtered comparison — which requires bubble dates identified independently via

GSADF — is designed to address that ambiguity. In our sample, the comparison moves the trace statistics without changing the inferred rank, so we report it as evidence of direction rather than magnitude.

Reviewer Comment:
The authors should provide a more detailed explanation of how the selected models complement one another and how each methodological component contributes to the overall analytical framework.

Author Response:
We agree and have added the following to the Introduction:

> The four steps are sequential rather than independent: GSADF-dated bubble episodes supply the exogenous filter used in the cointegration test, while the correlation network and cointegration results together motivate the comparison carried out via DCC-GARCH.

In short, the correlation network first maps how the sixteen series cluster within and across metals and identifies representative series; GSADF then dates bubble episodes series by series, a necessary input for the Johansen model's exogenous filter; and DCC-GARCH finally checks whether that long-run structure also holds day by day, as a correlation path rather than a single relationship estimated over the whole sample. This sequence lets us distinguish, for the same assets, structural market fragmentation from transient speculative distortion - a distinction no single method can draw on its own.

Reviewer Comment:
The contribution statement should explicitly demonstrate how the identified gaps in the literature are addressed, what previous studies have failed to uncover, and what novel evidence is provided by the present study.

Author Response:
We have substantially revised the contribution paragraph in the Introduction. Specifically, the third and fourth contributions have been reformulated to capture the novel evidence:

> This study makes several contributions to the understanding of critical-metal financialisation. First, we provide a comparative analysis of speculative bubble behaviour across all four metals using a unified GSADF framework applied to sixteen individual price series, addressing a gap in the literature that has largely focused on individual metals (Wang et al., 2023a; Peng et al., 2023; Restrepo et al., 2023b). Second, we gather multiple price series per metal from diverse sources to test the Law of One Price (Ardeni, 1989; Sarno et al., 2004; Bertone et al., 2015), extending Ardeni's (1989) cointegration approach by testing whether convergence strengthens once detected bubble episodes are filtered out. Third, we identify where price discovery resides within each metal: combining the VECM adjustment coefficients with formal tests of weak exogeneity, we find that Chinese exchanges lead price formation in every metal for which a leader can be identified, a pattern the raw estimates suggest but only the bubble-filtered model establishes. Fourth, we show that these dimensions of financialisation do not align: bubble propensity, long-run convergence

and short-run co-movement diverge across metals and even within them, so that financialisation cannot be captured by any single indicator.

Reviewer Comment:
The discussion of the findings should place greater emphasis on economic insights and policy implications, particularly in light of the acceptance or rejection of the proposed hypotheses.

Author Response:
We have substantially rewritten the Conclusions to address this directly, first summarising the result for each hypothesis in turn, then drawing out the broader economic insight:

> We find no significant association between series-level illiquidity and bubble intensity (H1), suggesting bubbles are driven more by idiosyncratic shocks than by market microstructure. Filtering out detected bubbles strengthens the evidence of cointegration for cobalt and nickel, in direction if not in magnitude (H2a); across metals, however, cointegration strength runs counter to financialisation (H2b). At the cross-commodity level, lithium and cobalt are clearly decoupled from copper and nickel (H3); within copper and nickel individually, however, short-run stability and long-run cointegration diverge, most plausibly reflecting the disruption around the March 2022 LME suspension rather than a difference in financialisation. Beyond the four hypotheses, the adjustment coefficients and weak-exogeneity tests locate price discovery in a Chinese venue in all three metals where a leader can be identified.
>
> Taken together, these results indicate that illiquidity, long-run convergence, and short-run co-movement behave as distinct - and at times contradictory - dimensions of integration that do not necessarily align even within the same metal: copper shows limited long-run cointegration among its own price series, yet the most stable short-run correlations in the sample. Financialisation, in other words, is not a single, unidimensional construct.

This is followed by a dedicated paragraph of practical implications for portfolio construction, risk management, and market analysis, each grounded in a specific result rather than stated generically.

Reviewer Comment:
The limitations of the study and directions for future research should be discussed in a more comprehensive and impactful manner.

Author Response:
We have added a substantially expanded discussion to the Conclusions:

> Several limitations should be noted. The DCC-GARCH analysis is restricted to copper and nickel, as the thin trading of lithium and cobalt introduces numerical singularities in multivariate GARCH estimation; our short-run evidence for these two metals is therefore limited to the cross-commodity comparison. The analysis covers a single four-year window rather than sub-samples, for reasons detailed in the Data and Methods section. We also do not find a data-supported explanation for lithium's high cointegration rank: its four

> price series combine two distinct product forms (carbonate and hydroxide) from a mix of sources, and disentangling this heterogeneity from financial arbitrage is beyond the present scope. We do not decompose bubble episodes by their macroeconomic or geopolitical drivers.
>
> These limitations point to future research directions. Building on the driver-decomposition approach of Peng et al. (2023) and Kangalli Uyar et al. (2024), future work could examine which macroeconomic and market-specific factors systematically predict bubble incidence across the sixteen series studied here. A second direction is to resolve the lithium anomaly noted above. Finally, as trading activity in lithium and cobalt deepens, extending the DCC-GARCH analysis to these metals - potentially with an asymmetric specification - would allow a fuller test of short-run co-movement across all four metals.

Reviewer Comment:
The authors should also provide a clearer justification for the decision not to conduct a sub-sample analysis.

Author Response:
We have added an explicit justification to the Data section:

> The sample period from 2021-07-19 to 2025-07-21 is the longest window over which daily data are simultaneously available across all sixteen series, bounded by the July 2021 launch of the lithium hydroxide instruments. The analysis conducted requires all series to be of equal length. This is why we fill missing observations using cubic spline interpolation, which leaves 1,042 daily returns for each series. A summary of all series and their sources is provided in Table 1. Splitting our sample is impractical: the sub-periods would be too short to estimate a four-variable DCC-GARCH system reliably, particularly for the thinly traded lithium and cobalt series, where the high incidence of zero returns reduces the effective sample further. We therefore treat the full sample as a single window and let the DCC estimates and the bubble dummy carry the time variation.

To put the constraint in numbers: the sample contains 1,042 daily returns, so a split into two sub-periods would leave roughly five hundred observations to estimate a four-variable DCC-GARCH system. For the lithium and cobalt series, where zero returns account for up to 88 per cent of trading days, the effective sample would be smaller still.

Reviewer Comment:
Overall, the manuscript addresses an interesting topic; however, its contribution would be substantially strengthened through a more theory-driven structure, clearer hypotheses, a stronger justification of the empirical strategy, and a broader discussion of the implications of the findings.

Author Response:
We thank the Reviewer again for these comments. Taken together, the changes outlined above — the four formal hypotheses, their explicit link to the literature gaps and empirical strategy, and the expanded discussion of results, limitations and future research — are directed at the Reviewer's overarching concern regarding theory-driven structure and empirical justification. We hope the Reviewer finds them responsive.

Reviewer #2

The idea of studying correlations, bubbles, and dynamic linkages in the critical metal markets is timely and draws important implications. However, I offer several comments to enhance the paper's quality, broaden its coverage of green and critical energy/metal markets literature, and strengthen the presentation and discussion of the results. The authors should carefully address the following comments.

Reviewer Comment:
Abstract should put more emphasis on the results and the implications of the findings.

Author Response:
We thank the Reviewer for this suggestion. The Abstract has been rewritten so that it now reports the four hypothesis-level findings in turn and closes on the implication we regard as most useful to practitioners: benchmarks for the same metal need not behave alike, so pricing reliability is better judged series-by-series than metal-by-metal. Working within the journal's guidance on abstract length, we have kept to this single overarching implication rather than enumerating several; the fuller discussion, covering diversification value, differentiated risk monitoring and the choice of reference price in long-term supply contracts, is developed in the Conclusions.

Reviewer Comment:
The introduction section is comprehensive; however, the authors should spend more effort to better highlight the motivation, and the contribution to the related academic literature. While doing this, they should better highlight why a four step methodological approach is necessary to serve the purpose of the study.

Author Response:
We have strengthened the motivation, added an explicit geopolitical rationale for the strategic importance of these metals, revised the contribution statement, and added a direct justification for the sequential four-step design (this last point is also raised by Reviewer 1; the same revision addresses both):

> Climate change and the global transition toward a net-zero economy increase the demand for some metals critical to this process, whose supply chains are increasingly entangled with geopolitical competition among major economic blocs (Kalantzakos, 2020), and make the pricing and financialisation of these new strategic assets ever more important.
>
> ...
>
> The four steps are sequential rather than independent: GSADF-dated bubble episodes supply the exogenous filter used in the cointegration test, while the correlation network and cointegration results together motivate the comparison carried out via DCC-GARCH.

The revised contribution paragraph (Introduction, fifth paragraph) now explicitly states the literature gap each contribution addresses; please see our response to Reviewer 1's fifth comment for the full text.

Reviewer Comment:
The literature review should be extended to cover the growing evidence on the impact of

bubbles in the commodity (green energy and metal) markets. This would improve the positioning of your analysis and make it more susceptible to potential citations in the field. Accordingly, the following related papers under the transition to the zero carbon economy are relevant and should be considered while positioning your paper:

- *"Price Bubbles and Co-bubbles in the Green Economy Market." Journal of Environmental Management, 370, 122678.*
- *"The race for critical minerals in an era of geopolitical realignments." The International Spectator, 55(3), 1-16.*
- *"Co-bubble transmission and portfolio implications: A dynamic network approach covering two crisis periods and clean versus dirty cryptocurrencies." Journal of International Money and Finance, 145, 103108.*
- *"Climate Risk and the Volatility of Agricultural Commodity Price Fluctuations: A Prediction Experiment." In Behavioral Finance and Asset Prices: The Influence of Investor's Emotions (pp. 23-44). Cham: Springer International Publishing.*
- *"Green Minerals and Energy Markets in the Low-Carbon Transition: Time-Frequency Connectedness and Portfolio Implications." Mineral Economics, https://doi.org/10.1007/s13563-026-00602-5.*
- *"Rare earth and allied sectors in stock markets: Extreme dependence of return and volatility." Applied Economics, 53 (49), 5710-5730.*
- *"The battery bubble." Journal of Environmental Planning and Management, 1-28.*
- *"Dependence structure among rare earth and financial markets: A multiscale-vine copula approach." Resources Policy, 83, 103626.*
- *"Energy consumption, finance, and climate change: does policy uncertainty matter?." Economic Analysis and Policy, Vol. 70, 490-501.*

Author Response:
We are grateful for these highly relevant suggestions. We have rewritten the Literature Review section integrating all nine recommended papers directly into it.

We are grateful especially for Gupta and Pierdzioch (2023) and Atsu and Adams (2021) papers, which have helped us frame the non-fundamental component of critical-metal pricing more broadly than we had initially.

Reviewer Comment:
Data and methods: Justify the choice of the sample period and importantly the various indices used.

- *Define more clearly the critical metals used in the analysis. Are the indices liquid enough?*
- *Why not using an asymmetric DCC process?*

Author Response:
We address each of the points in turn. On the sample period:

> The sample period from 2021-07-19 to 2025-07-21 is the longest window over which daily data are simultaneously available across all sixteen series, bounded by the July 2021 launch of the lithium hydroxide instruments.

On the definition of the metals and the indices, the Data section now introduces each venue and information provider before any series is named, states the four series used for each metal, and specifies the product form where it differs within a metal – lithium, whose series combine carbonate and hydroxide; the note to Table 1 defines every ticker and its source.

On whether the indices are liquid enough, we have tried to answer this from the data rather than assume it. Table 2 now reports the incidence of zero-return days for all sixteen series, from below 0.1% for the daily copper and nickel benchmarks to 87.9% for the COMEX cobalt series, so that adequacy can be judged series by series rather than taken on trust. We show that series-level liquidity varies across sources even within the same metal.

Our answer is accordingly method-specific. For the analyses conducted on daily log price levels – the correlation network, the GSADF tests and the Johansen system – all sixteen series are available over a common window of 1,042 daily returns, and a repeated quote enters as an unchanged price rather than as a missing observation; we regard them as adequate there, if less precisely estimated for the thinnest series. For multivariate GARCH they are not, and we have allowed that to bind: the within-metal DCC systems for lithium and cobalt fail to converge and are not reported, and we do not split the sample. We would add that this heterogeneity is also part of what the paper measures – H1 tests it directly and H3 rests on it – so restricting the analysis to the most liquid benchmarks would remove the variation the hypotheses are designed to exploit.

On the choice of a symmetric DCC specification:

> We employ the symmetric DCC specification of Engle (2002) rather than an asymmetric variant (e.g., Cappiello et al., 2006). Given the limited number of trading days available for the lithium and cobalt series, the additional parameters required by an asymmetric specification would further strain model convergence.

Reviewer Comment:

Results: I would have liked to see a more detailed discussion of results, while putting more emphasis on economic institutions and discussions of results in light of previous findings on the importance of green energy and critical metal markets. This would add more value to the analyses, making it more susceptible to citations from the rising literature on the topic.

Author Response:

We have expanded the discussion of results throughout the Empirical Results section and the Conclusions to connect our findings more explicitly to the existing literature, for instance linking the 2022 nickel episode to both Wang et al. (2023b) and Vuong et al. (2025), and framing the overall pattern of results in terms of the broader financialisation literature. Please see our response to Reviewer 1's sixth comment for the relevant Conclusions text.

Reviewer Comment:

Try to be more careful in the presentation of results for Figure 4.

- *Add more explanations about the following: "... battery metals sector has not yet integrated into the traditional commodity market".*
- *Are you using explosive and bubble interchangeably?*

Author Response:
We believe this comment refers to Figure 5 (the cross-commodity DCC figure, immediately preceding the Conclusions), where this sentence appears; Figure 4 shows only the within-metal nickel correlations. We have expanded the surrounding discussion and clarified our terminology. On the battery-metals finding, the relevant paragraph now reads:

> Pricing in these two markets appears driven by idiosyncratic, localised factors such as EV subsidy shifts rather than by systemic global trends, and from a portfolio perspective this orthogonality implies substantial diversification value against traditional industrial commodities. The disconnection is, however, specific to the industrial-metals complex: Malhotra et al. (2026) find that lithium becomes a transmitter of volatility spillovers to energy, precious-metal and equity markets during severe shocks, so these near-zero correlations against copper and nickel should not be read as isolation from financial markets more generally.

We are grateful for the remark: the original phrasing generalised beyond what a four-asset cross-commodity system can support.
On terminology, we have added the following clarification where the GSADF test is first introduced in Methods:

> Throughout the paper, we use the terms "explosive behaviour" and "bubble" interchangeably: the former is the statistical property tested by the GSADF procedure, while the latter is the economic label conventionally applied to the resulting dated episodes (Phillips et al., 2011, 2015).

Reviewer Comment:
In the conclusion section, try not overstating the findings from the analysis. In other words, keep your results within the framework of the methodology. Furthermore, provide a more detailed discussion of the implications would make the paper richer and more informative. However, be careful not to overstate the findings while presenting the implications.

Author Response:
We have substantially rewritten the Conclusions with this balance in mind: each implication is now tied to a specific, hedged result rather than stated as a general claim. For example:

> The short-run decoupling of lithium and cobalt confirms their diversification value against more integrated metals, though this reflects thin trading rather than a stable structural feature and may narrow as these markets mature. The 2022 nickel episode illustrates that episodic, event-driven risk is distinct from the structural illiquidity risk in lithium and cobalt, warranting different

monitoring approaches. More broadly, our finding that cointegration strength does not reliably track financialisation across metals suggests that single-indicator assessments of a metal's "financialisation level" should be treated with caution, and that reliable pricing sources are better identified series-by-series than metal-by-metal.

Reviewer Comment:
Make sure that all cited papers are presented in the reference list and vice versa.

Author Response:
Thank you for this remark. We have conducted a full cross-check of in-text citations against the reference list in both directions and identified and corrected several issues: a missing reference entry for Peng et al. (2023), an orphaned reference (Będowska-Sójka, 2021) whose in-text citation had been inadvertently dropped from the liquidity discussion and has now been restored, disambiguation of two distinct Wang et al. (2023) papers as 2023a (lithium) and 2023b (nickel) throughout the text and reference list, and correction of the alphabetical ordering of several entries.

Reviewer Comment:
Make your tables and figures self-explanatory.

Author Response:
We have reviewed and expanded the note to every table and figure; Table 1's note already carried the full source and ticker definitions and required no addition.

Reviewer #3

I appreciate the authors' revision and response. The main concern in the previous round was the quality and readability of the figures. The revised manuscript has improved substantially in this respect. Figure 1 is now clearer, the links are more visible, and Figures 2-5 are easier to read than before. Overall, the authors have addressed the major technical concern raised in the previous review. I have only a few remaining minor comments:

Reviewer Comment:
1. Figure 1 caption should be corrected. The note currently says, "Each node represents a given metal." This is not precise. Each node appears to represent a specific price series or benchmark for a metal, not the metal itself. Please revise the note accordingly.

Author Response:
We thank the Reviewer for catching this imprecision. The note has been corrected:

> Note: Each node represents an individual price series for one of the four metals, distinguished by colour: orange for copper, green for nickel, grey for lithium, and blue for cobalt. Series abbreviations are defined in Table 1.

Reviewer Comment:
2. Figure 1 could still be improved slightly. Although the red links are now more visible, several labels overlap with nodes or edges. If possible, please slightly adjust label placement or increase the figure size to improve readability.

Author Response:
Thank you for pointing this out. Figure 1 has been regenerated from the source code with adjusted label placement and larger figure dimensions to minimise the overlap between labels, nodes, and edges. We have also colour-coded the nodes by metal, as recorded in the note quoted above, so that the four groups of series can be distinguished at a glance rather than read off the labels. We hope the figure is now easier to read.

Reviewer Comment:
3. Figure notes should be made more informative. For Figures 3-5, the notes only refer readers to Table 1 for abbreviations. It would be helpful to add one sentence explaining that the plotted values are estimated dynamic conditional correlations and that higher values indicate stronger time-varying co-movement between the corresponding series.

Author Response:
We agree and have added the suggested clarification to the notes of Figures 3, 4, and 5, each naming the series it plots. The note to Figure 3, for example, now reads:

> Note: Plotted values are the estimated dynamic conditional correlations (DCC) for all six pairs formed from the four copper price series over the sample period from 2021-07-19 to 2025-07-21; higher values indicate stronger time-varying co-movement. Series abbreviations and data sources are defined in Table 1.

Reviewer Comment:
4. Please ensure that each table and figure has a clear caption/title and, where appropriate, brief explanatory notes so that readers can easily understand the content. In

particular, tables reporting bubble episodes, cointegration tests, and DCC results should clearly define the variables, test statistics, and interpretation of each reported column.

Author Response:
We have reviewed the caption and note of every table and figure with this requirement in mind. Each note now defines the reported columns, states the test statistic and its null hypothesis, and gives the interpretation of the reported coefficients.

For the three items the Reviewer names specifically: the note to Table 3 states the unit of observation and defines both reported columns – the total number of days classified as a bubble and the number of distinct episodes – over the stated sample period; the note to Table 4 defines the null and alternative of the trace test, the significance conventions, and the basis on which the price follower and leader are identified, including the weak exogeneity test now reported alongside the adjustment coefficients; and for the DCC results, the estimated parameters, their significance and the implied persistence are now reported at the start of the Dynamic Conditional Correlation Models subsection, while the notes to Figures 3–5 state what the plotted values represent and over what period.

In addressing this remark, we re-examined the code underlying Table 4 and identified an error in the extraction of the VECM adjustment coefficients: for metals with a cointegration rank greater than one, the reported values corresponded to the loadings of successive cointegrating vectors within a single equation rather than to the adjustment speeds of the four price series. Cobalt, lithium and nickel were affected; copper, estimated at rank one, was not. Table 4 now reports, for each series, the Euclidean norm of its row of the adjustment matrix, together with a likelihood-ratio test of weak exogeneity. The cointegration ranks and trace statistics are unchanged, and neither H2a nor H2b is assessed on the adjustment coefficients, so the conclusions drawn in the revised manuscript are unaffected. The identification of price leaders and followers has been revised accordingly, and the corresponding discussion rewritten.

Reviewer Comment:
5. There is room to improve the literature review. Please add a brief discussion of recent related studies such as Guesmi et al. (2026) on quantile-frequency connectedness among green assets, crude oil, gold, and stock markets; Li et al. (2025) on precious metals as hedges for clean energy stocks; and Fang et al. (2025) on oil shocks, clean energy dynamics, and automaker stock performance. These studies are closely related to the manuscript's focus on critical metals, commodity-market financialisation, energy-transition assets, connectedness, hedging, and portfolio implications.

Author Response:
We thank the Reviewer for these suggestions and have incorporated all three studies into a new paragraph in the Literature Review situating our work within the broader green-asset connectedness literature:

> This literature connects to broader work on the interconnectedness of green and commodity markets, including co-bubble transmission across clean and dirty cryptocurrency markets (Chen et al., 2024), rare earth elements' dependence on and connectedness with broader financial markets (Bouri et al., 2021; Kamal and Bouri, 2023), and quantile-frequency connectedness between green assets and traditional commodities such as crude oil and gold (Guesmi et al., 2026). Precious metals have similarly been studied as potential hedges for clean energy equities (Li et al., 2025). Fang et al. (2025) show that oil-market

> shocks and clean-energy dynamics transmit asymmetrically to traditional versus electric-vehicle automaker equities. The demand side of the battery-metal chain is therefore already financially integrated, while our results suggest the underlying metal markets are not.

We found Fang et al. (2025) a useful counterpoint to our own results, since it suggests that financial integration along the battery-metal value chain may be established downstream before it is established in the metal markets themselves.

Reviewer Comment:
6. Carefully check and update the reference list. Please ensure that all citations, including newly added references, are complete, consistent, and correctly formatted according to the journal's style.

Author Response:
We have conducted a full cross-check of the reference list against in-text citations in both directions. This identified and corrected: a missing entry for Peng et al. (2023); an orphaned entry (Będowska-Sójka, 2021) whose in-text citation had been dropped and is now restored in the discussion of the ZeroPct trading inactivity proxy; incomplete bibliographic details for Wang et al. (2023b), now supplemented with volume and article number; disambiguation of two distinct Wang et al. (2023) papers as 2023a (lithium) and 2023b (nickel) throughout; and several alphabetical ordering errors. All newly added references have been formatted consistently with the existing list.

Reviewer Comment:
7. Final proofreading is recommended. The manuscript is generally readable, but a few small wording and punctuation issues remain. For example, some reference parentheses appear incomplete in the literature review, and some sentences could be polished for clarity.

Author Response:
We are grateful for this comment. A full proofreading pass has been completed. Among other corrections, we identified and fixed four incomplete citation parentheses in the Literature Review (in the citations to Szafranek et al., 2023; Basak and Pavlova, 2016; Bertone et al., 2015; and Ozgur et al., 2021), a missing citation to Engle (2002) in the description of the DCC-GARCH model, an ungrammatical sentence fragment in the discussion of the Law of One Price, and several residual formatting artefacts. We have also polished a number of sentences throughout for clarity and concision.

Reviewer Comment:
Overall, the authors have satisfactorily addressed the previous concern. I recommend minor revision, mainly for final figure-caption correction, figure-note clarification, literature positioning, reference-list checking, and proofreading.

A note on manuscript length

In addressing the Reviewers' comments, the manuscript has grown beyond the journal's limit of 7,500 words. The additions are a direct consequence of the revisions requested: four formal hypotheses with their theoretical development, newly integrated references, expanded justifications in the Data and Methods section, a fuller discussion of results in

light of each hypothesis, and a new discussion of limitations and future research. We have offset these additions wherever possible by condensing descriptive passages throughout the manuscript, and would be glad to shorten the text further should the Editor consider it necessary.